

FINAL PROJECT

1. Objectives

The goals of this assignment are to use a popular machine learning tool, i.e. scikit-learn, to investigate two important factors for the success of machine learning applications, which are performance metrics and optimisation. The specific objectives of this assignment are:

- Be able to use different classification/regression methods implemented in scikit-learn, such as kNN, support vector machines, decision tree, random forest, AdaBoost, gradient boosting, linear discriminant analysis, logistic regression, linear regression, k-neighbors regression, Ridge regression, decision tree regression, random forest regression, gradient Boosting regression.
- Compare the performance of different classification/regression methods using a number of popular performance metrics, e.g., accuracy, precision, recall, confusion matrix, AUC under ROC, ... and analyse the results.
- Compare and analyse advantages and disadvantages based on the results of different performance metrics for a given regression task.

Table 1. Dataset List for Project

No	Dataset	Loại bài toán
1	Bank Note Authentication (banknote_authentication.csv)	Binary Classification
2	Horse Colic (horse-colic.csv)	Binary Classification
3	Ionosphere (ionosphere.csv)	Binary Classification
4	Sonar Returns (sonar.csv)	Binary Classification
5	German Credit (german.csv)	Binary Classification
6	Oil Spill (oil-spill.csv)	Binary Classification
7	Phoneme (phoneme.csv)	Binary Classification
8	Thyroid Gland (new-thyroid.csv)	Multiclass Classification
9	Ecoli (ecoli.csv)	Multiclass Classification
10	Glass Identification (glass.csv)	Multiclass Classification
11	Wheat Seeds (wheat-seeds.csv)	Multiclass Classification
12	Abalone Age (abalone.csv)	Regression
13	Wine Quality Red (winequality-red.csv)	Regression
14	Auto Insurance Total Claims (auto-insurance.csv)	Regression
15	Auto Imports Prices (auto_imports.csv)	Regression

* Each group registries to work on different datasets.

The given dataset list above, the task here is to learn a regression (classification) model to discover the relationship between the output variable and the input variables.

2. Instructions

As we discussed in the lectures/tutorials, to use scikit-learn for regression/classification, you may need the following seven steps:

- Step 1. Load Data
- Step 2. Initial Data Analysis
- Step 3. Exploratory Data Analysis
- Step 4. Preprocess Data
- Step 5. Build classification (or regression) models using the training data
- Step 6. Evaluate models by using cross validation (Optional)
- Step 7. Assess model on the test data.

Detailed step-by-step instructions are given below. Please make sure you read them thoroughly.

- **Step 1: Conduct exploratory data analysis (EDA).**

By performing EDA, you can have a better understanding about the data, which helps you decide whether to perform pre-processing (such as image filtering or feature engineering) and which methods to perform in order to improve the data quality. Check they're all the right size.

- **Step 2: Apply pre-processing techniques to improve quality of the data.**

For simple pre-processing, you can just simply remove some noisy images/instances. Alternatively, you could apply feature engineering techniques (e.g. image feature descriptors to extract high-level features, feature extraction, or feature constructions) to try and improve the quality of the data and hence classification accuracy of your final model.

- **Step 3: Build a simple baseline model**

Experiment Setting, use “42” as the random seed to split the data into a training set and a test set, with 70% as the training data and 30% as the test data. In training phase, use the k-fold strategy with $k=5$.

For classification problem, you should use 10 classification algorithms implemented in scikit-learn to perform classification. These 10 algorithms are very popular classification methods from different paradigms of machine learning: (1) kNN, (2) naive Bayes, (3) SVM, (4) decision tree, (5) random forest, (6) AdaBoost, (7) gradient Boosting, (8) linear discriminant analysis, (9) multi-layer perceptron, and (10) logistic regression. You are encouraged to read the documentation (and provided references if you would like to know more details) about these methods from scikit-learn, e.g. kNN is implemented in `sklearn.neighbors.KNeighborsClassifier`.

For regression problem, you should use the following 10 regression algorithms implemented in scikit-learn to perform regression. These 10 algorithms are very popular regression methods: (1) linear regression, (2) k-neighbors regression, (3) Ridge regression, (4) decision tree regression, (5) random forest regression, (6) gradient Boosting regression, (7) SGD regression, (8) support vector regression (SVR), (9) linear SVR, and (10) multi-layer perceptron regression. You are encouraged to read the documentation (and provided references if you would like to know more details) about these methods from scikit-learn, e.g. linear regression is implemented in `sklearn.linear_model.LinearRegression`.

- **Step 4: Discussion about results**

For the classification results:

- Based on exploratory data analysis, discuss what preprocessing that you need to do before classification, and provide evidence and justifications.
- Report the results (keep 2 decimals) of all the 10 classification algorithms on the given test data in terms of classification accuracy, precision, recall, F1-score, and AUC. You should report them in a table.
- Find the two best algorithms according to each of the four performance metrics, Are they the same? Explain why.

For the regression results:

- Based on exploratory data analysis, discuss what preprocessing that you need to do before regression, and provide evidence and justifications.
- Report the results (keep 2 decimals) of all the 10 regression algorithms on the test data in terms of mean squared error (MSE), root mean squared error (RMSE), relative squared error (RSE), mean absolute error (MAE), and execution time. You should report them in a table.
- Compare the performance of different regression algorithms in terms of MSE, RMSE, RSE, and MAE, then analyse and discuss their differences and provide conclusions.

- **Step 5: Tuning the models**

You should tweak the model built from Step 3 by using the knowledge you have learned so far. Some suggestions are given below and you should try at least five of the following points:

1. Investigate hyper-parameters of every model.
2. Use cross-validation on the given datasets to tune the model.
3. Use ensemble learning (e.g., stacking different models) to boost your model.
4. Some other option of your own choosing!

- **Step 5: Write a report.**

For this step, you are required to complete a formal written report. Please check the detailed task description and submission requirements below.

3. Task Description and Submission Requirements

Report.docx: You must submit a written report in Word Format, describing the methodology you used and the results. The report should include the following information, and it probably makes sense to make these section headings as it helps the reader to navigate.

- **Introduction:** briefly describe the problem and summarise your approach.
- **Problem investigation:** describe and justify what you have done in terms of EDA, data pre-processing, feature design/selection methods, and any other methods/algorithms you used to improve your data. A good way to report this part is to describe your findings in EDA, the (pre-)processing you have done based on the findings (if applicable), why you did such processing, and whether such processing helped improve the model.
- **Experiments and Discussion**

For experimental setup, you are required describe and justify what you have done for training the models in terms of the following aspects if applicable:

1. how you use the given dataset (e.g. how you split them into training and validation sets or k-fold cross validation, and how you use them for training),
2. hyper-parameter settings,
3. technical advancements you have made to improve your model.

For result and and discussion, this should take the form of a self-contained summary, in which you clearly and succinctly:

- describe the structure and overall settings of your chosen baseline methods, and the best method.
- compare the results in terms of the training time and the classification (regression) performance (for validation/test data). Plots of loss and accuracy versus time, for train and test sets, would be useful graphics to show.
- note any striking differences, and discuss why you think they occurred.

- **Conclusions and future work**

Describe the overall conclusion(s) that can be drawn from your study, show insight into the potential pros and cons of whatever approach you took, and list next steps / future work. that you would suggest could be undertaken to take things further. Aim to exhibit insight and understanding here - simply saying 'do more of the obvious things' for next steps doesn't amount to insight :) (note: it's not a 'Reflection' about your own experience, but a technical conclusion to a technical study)

Present.pptx: preparing the slide for presentation with the parts as the report: Introduction, Problem Investigation, Experiments and Discussion, Conclusions and Futuer Work.

Code Folder: cointaing the code and experiment results relating to your project.

Readme.docx: writing the information about student name, student id, and roles.

All materials should be compress and named such as: **SGU-[GroupNo]-[Dataset Name]-Final.zip** and upload to LMS.

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(End)